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Python II
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Pittsburgh 311 Data Exploration

Summary
This project seeks to find interesting relationships in a large dataset from the City of Pittsburgh that tracks requests made through its 311 customer relations management system. Using the dataset, I hope to identify types of 311 requests that are most common, and types of requests that are time consuming to address. The results of this analysis will provide some insight into how the city can further prepare different departments to handle citizen requests effectively and verify an equitable and consistent application of public services over different locations within Pittsburgh and modes of communication. The analysis will also offer interesting information to Pittsburgh residents about how the city works for citizens and how long one should expect to wait for different types of 311 service requests to be completed.

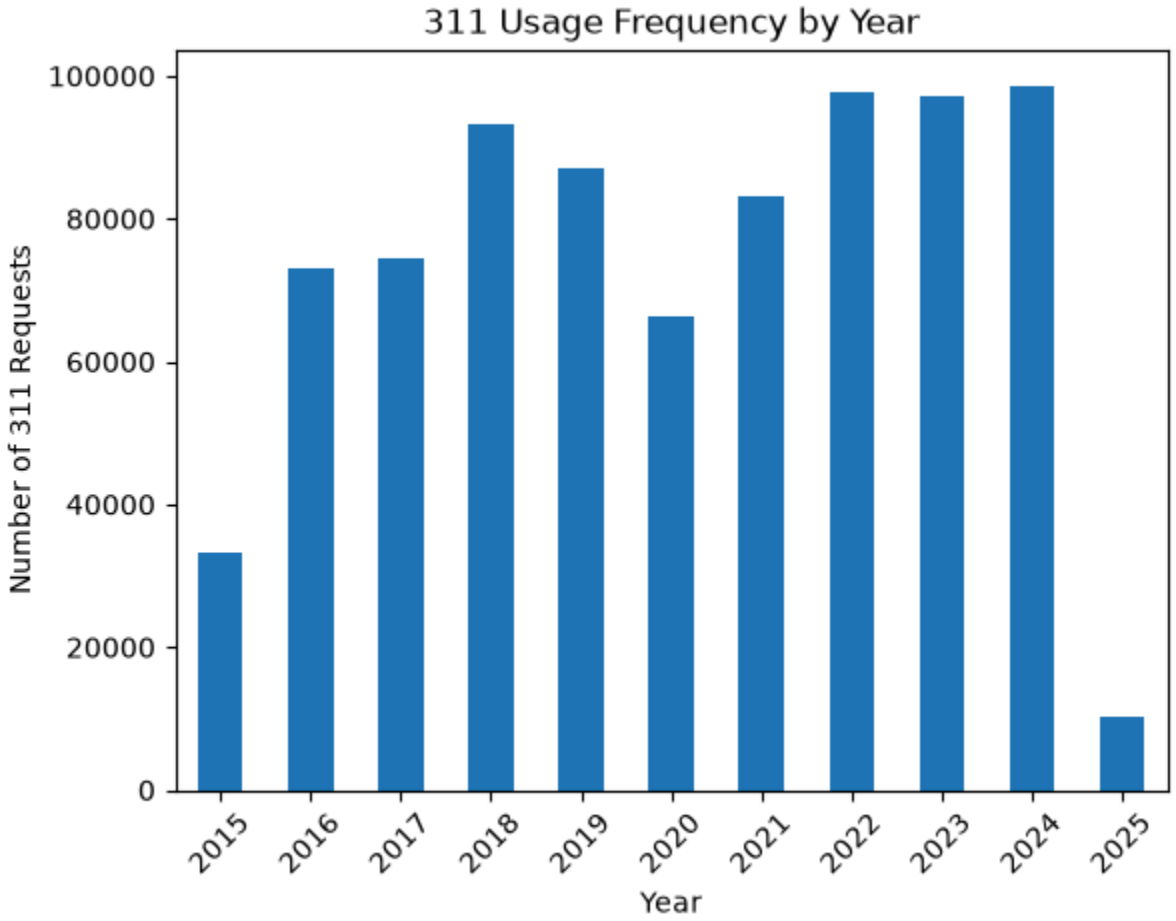
311 Background and Data Introduction

The 311 system exists so anyone in the city can contact an official body of the city and request service or get information about the city of Pittsburgh. The system is referred to as a 311 system because traditionally, contact to make a request would be made by dialling 311 on the phone. Today, Pittsburgh's system connects ten distinct modes of connection, including phone, city website, email, text, and a variety of apps. The following table shows the distribution of request origins in the dataset used here.

origin	count	percent
Call Center	482584	59.18%
Website	200158	24.55%
Control Panel	56489	6.93%
Report2Gov iOS	30187	3.7%
Twitter	12893	1.58%
Report2Gov Android	10633	1.3%
Report2Gov Website	9660	1.18%
Email	8374	1.03%
Text Message	4434	0.54%
QAlert Mobile iOS	5	0.0%

The table above shows that while the system has adapted to take requests from additional modes of communication, phone calls, represented by “Call Center” still represent the majority of requests.

The data used in this project is from the 311 system infrastructure used from April 20th, 2015 until March 10th, 2025. Usage of the system over time has low variance as seen in the following chart:

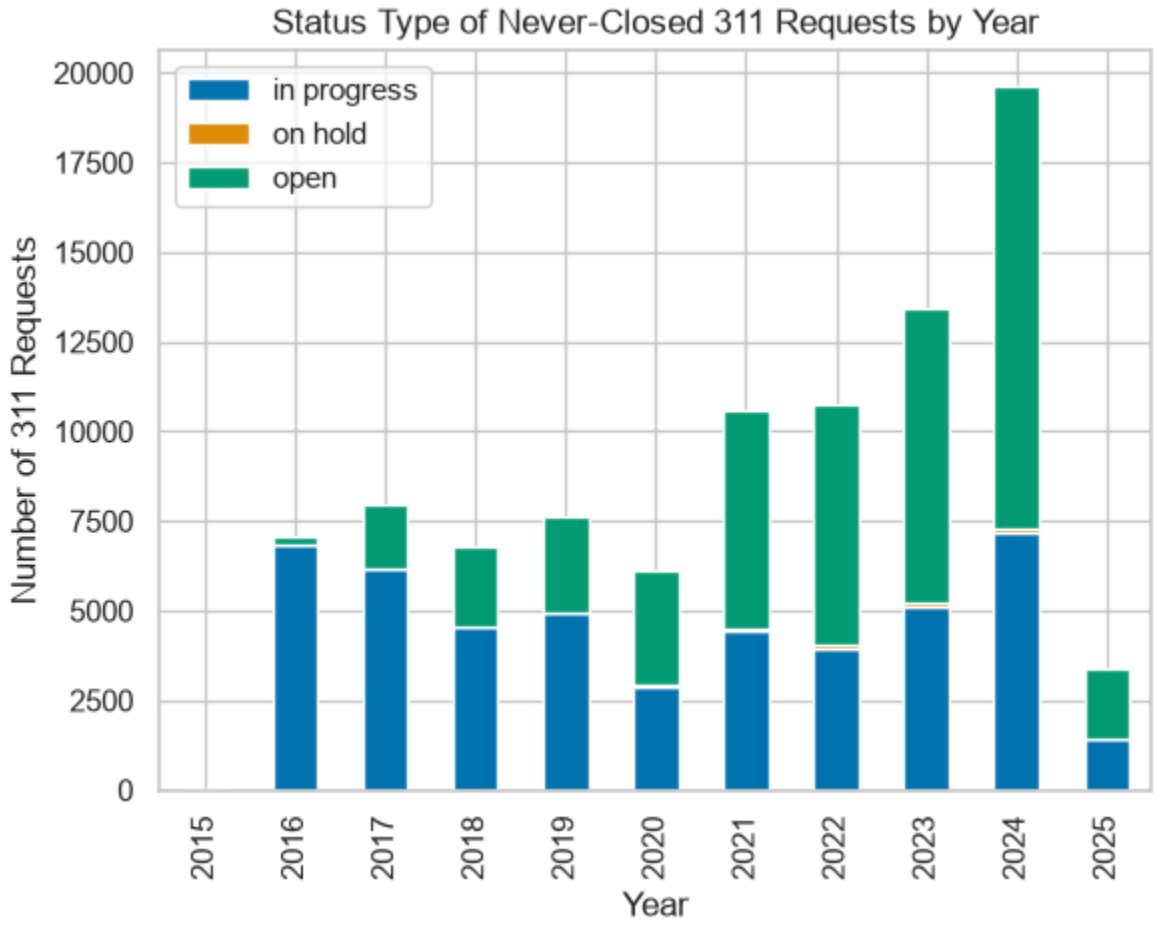


The first and last years that appear in the dataset have fewer overall requests because they do not encompass the entire year’s worth of requests. 2025 only has about three months worth of requests, and 2015 only contains about eight months of data rather than an entire year. Disregarding these two years, the overall usage data appears relatively flat, with a small dip in 2020 that might be attributable to the Covid-19 pandemic

The approximately 10 years of data consists of 81,4330 requests (rows) and 32 distinct variables (columns). The complete list of variables used in this analysis can be found in the associated data dictionary along with any dataframes created during analysis.

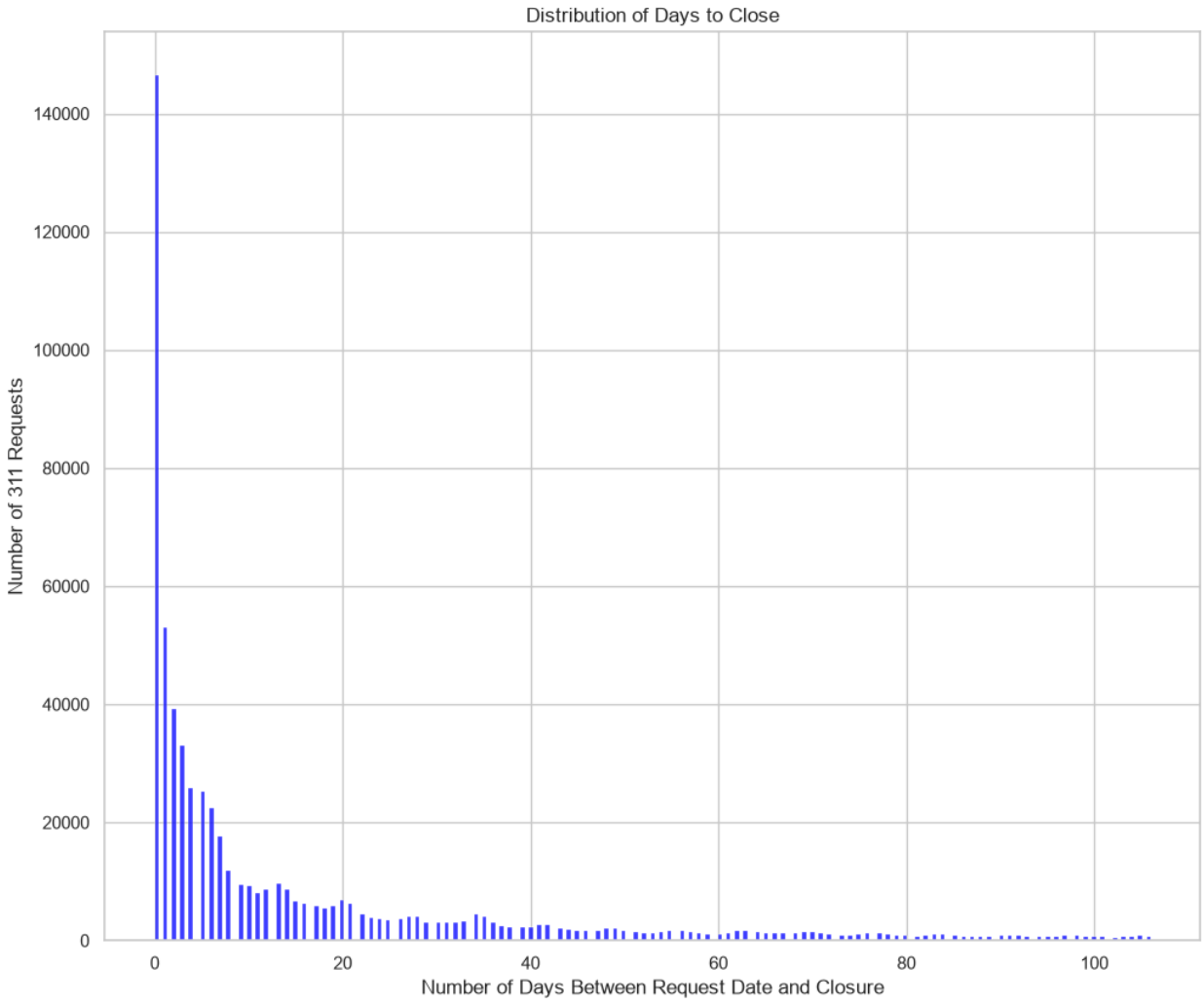
Analysis of 311 Request Duration

The primary variable of interest for this analysis is days_to_close, the duration (in days) that a 311 service request remains open before it is closed. This variable is generated by taking the difference between closed_date_et and create_date_et. Since not all of the requests were closed in the dataset, this means days_to_close is missing 93,338 cases. The distribution of never-closed cases by start_date can be seen below, organized by status:



As one might expect, the number of never-closed requests rises as the year gets closer to the scope of the dataset. This is because the scope of the dataset cannot cover the entire duration of every request as the year moves toward the end of the data. Also note that the number of “open” requests increase relative to the “in progress” requests for almost every consecutive year, showing a positive relationship between age of a request and its likely status.

For 311 requests where the closing date fits within the dataset, the distribution of days_to_close can be seen below:



The x axis on the above chart is truncated to exclude outliers, which fall above 106 days using the 1.5 IQR method. The complete distribution of days_to_close is very wide, with a minimum of 0 and a maximum of 3,529 days. The mean of this variable is 86.8 and the median is 7, further showing an extreme rightward skew. This means that extremely time consuming requests are relatively rare among requests that were closed.

This data contains a variable called “request_type_name” which serves as a standardized categorical bucket for 311 requests. There are a total of 327 different categories under this

variable. For the sake of displaying data, I will mostly cover the most common categories. Below are tables of the ten most and least common 311 requests and their relative frequency.

Analysis of request_type_name

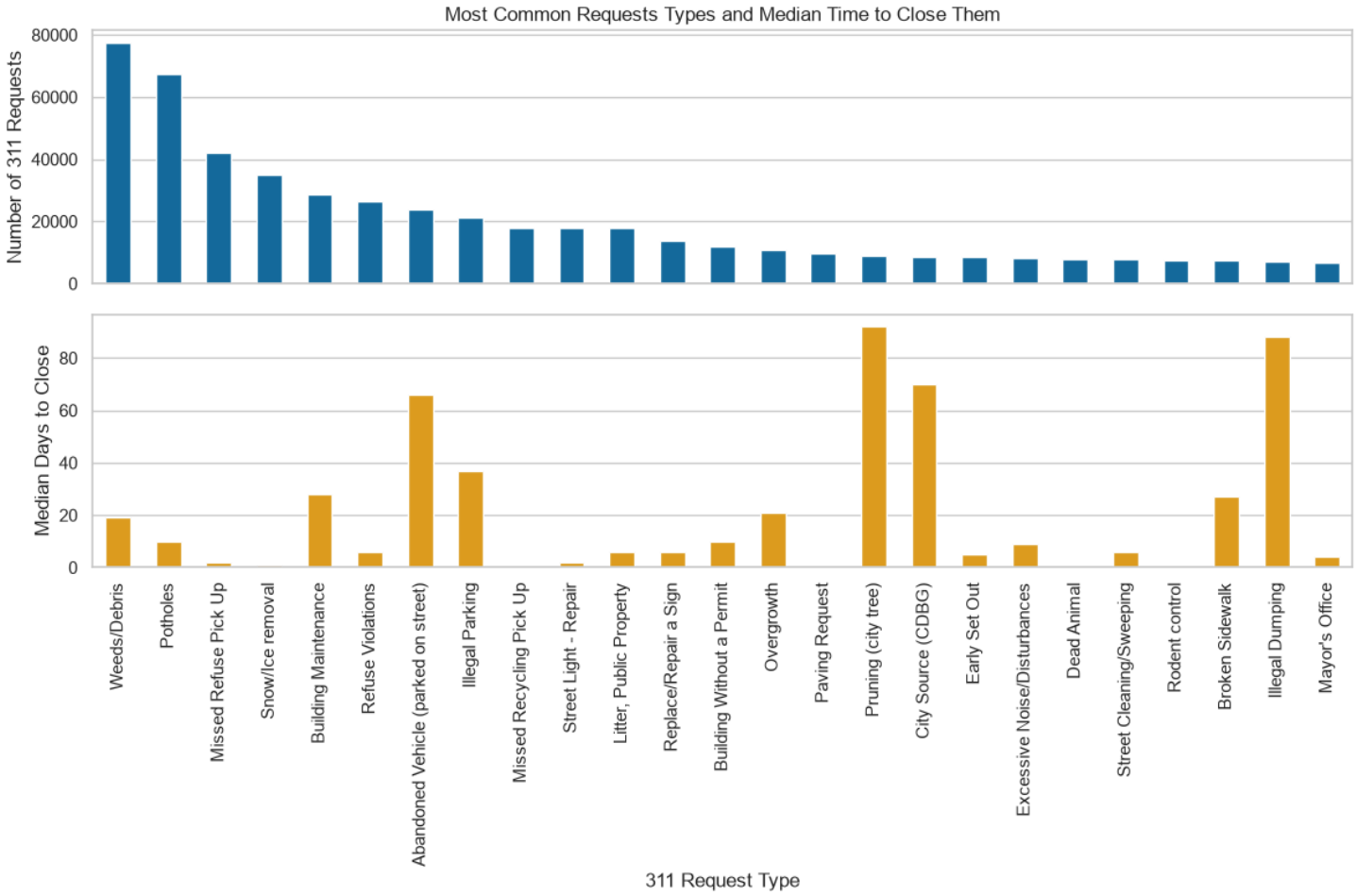
10 Most Frequent 311 Request Types:

request_type_name	count	percent
Weeds/Debris	77513	9.52%
Potholes	67502	8.29%
Missed Refuse Pick Up	42185	5.18%
Snow/Ice removal	34942	4.29%
Building Maintenance	28863	3.54%
Refuse Violations	26598	3.27%
Abandoned Vehicle (parked on street)	23768	2.92%
Illegal Parking	21289	2.61%
Missed Recycling Pick Up	17987	2.21%
Street Light - Repair	17903	2.2%

10 Least Frequent 311 Request Types:

request_type_name	count	percent
EMT	7	0.0%
KNOWLEDGE BASE ARTICLES	7	0.0%
Improvements	6	0.0%
Land Bank	5	0.0%
Bicycle/Pedestrian/Trail - Network Improvements	3	0.0%
Catch Basins	3	0.0%
Billing	3	0.0%
Permits	3	0.0%
Roadway Obstructions	1	0.0%
Amenities	1	0.0%

The following shows the 25 most common 311 requests in graphical form above the median time to close for each type.



This chart shows the 25 most common request types and the median number of days to close them. For example, the median number of days for the city to close 311 requests related to potholes is only 10 days. This chart effectively shows how long one might expect to wait for service based only on the type of request being made.

The following correlation table shows pairwise correlation between days_to_close and dummy variables for each request type. The magnitude of the correlation coefficients are essentially negligible, suggesting that there are other more relevant factors that better explain how long a 311 request remains open. This also reflects the large quantity of request_type categories.

Correlation between days to close and request_type dummies.

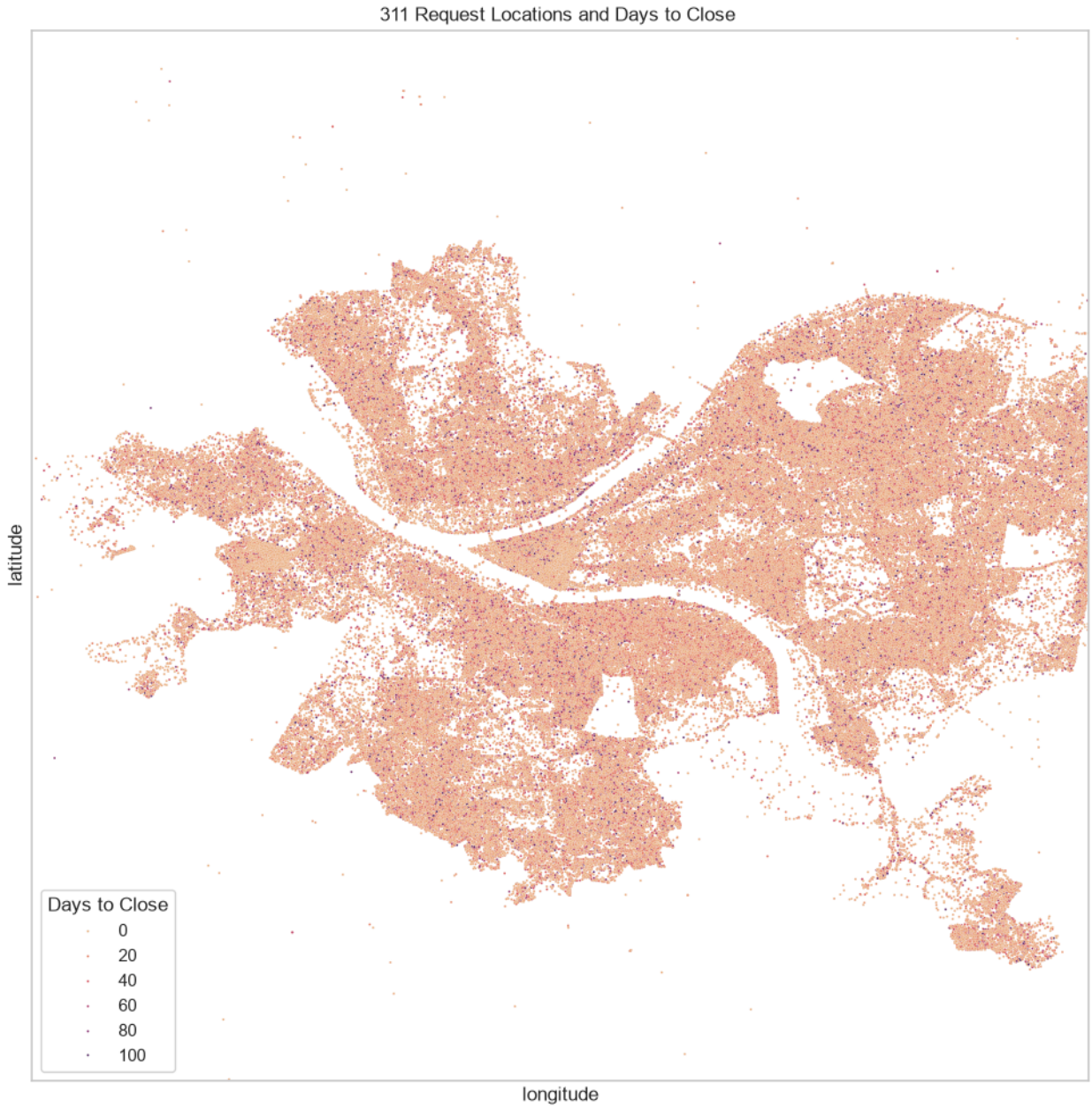
Request Type	**Correlation With Respect to Days to Close**

Sidewalk/Curb/ADA Ramp Maintenance	0.151204
Dead Tree (3TB)	0.087712
Repair City Steps	0.087234
Blocked or Closed Sidewalks	0.085644
Homeless	0.081577
Illegal Parking	0.079590
Traffic Enforcement	0.074510
Pruning (city tree)	0.068901
Patrol	0.067842
Speeding	0.062433
Retaining Wall (Public)	0.062379
Vacant Building	0.057738
Drug Enforcement	0.056124
Abandoned Vehicle (parked on street)	0.055353
Guide Rail	0.049576
County Maintenance	0.044030
Mayor's Office	0.043998
Illegal Dumping	0.040738
Water Runoff in ROW	0.040715
Brick or Block Repair	0.039578
Building Maintenance	0.038792
Traffic Shop	0.036224
Excessive Noise/Disturbances	0.034404
Graffiti, Documentation	0.033950
Boat/Trailer on Street	0.033652

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Location Data Exploration

As a brief exploration of location data and days to close, I plotted 311 latitude/longitude on a scatter plot and set the color scale to reflect days_to_close. I also filtered out outliers for the days_to_close variable to avoid skewing the color scale and making it more difficult to interpret. If there were any significant location patterns, one might expect to see clusters of darker or lighter points on the map in one particular location or area. This is essentially a GIS technique without a map layer, however the density of 311 request data is so high that the map of Pittsburgh emerges from the location data.



The result of this plot could reasonably represent a first view on whether there are any location-related biases on the speed at which the city of Pittsburgh addresses 311 requests by residents across the entire dataset analyzed here.

Conclusion

Based on the research question, this research was only able to take a basic first pass on the dataset. However, the median days_to_close for the most common 311 requests shows some variance in the time it takes for the city of Pittsburgh to address common 311 calls. Finally, a simple visualization of location data suggests that location across the city does not have any bearing on time_to_close.